

Design and Function

Clutch

General

For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components," publication number 175-001, and appropriate vendor literature.

Hydraulic fluid in the clutch system collects moisture from the air and will eventually hold enough moisture to affect the metal surfaces in the system unless removed. Replace the fluid at the recommended intervals or more frequently.

The clutch pedal play (1) is given by the clearance between the plunger and the piston (1a) in the master cylinder. Thus the pedal will always have a play, regardless of the clutch adjustment. The correct play is adjusted with the upper adjusting screw (1b) in the pedal carrier.

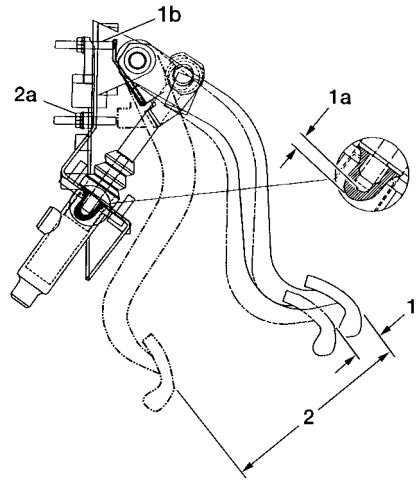


Fig. 1: Clutch

Volvo Clutch Slave Cylinder

Slave cylinder stroke (A) is 29 ± 1 mm (1.14 ± 0.04 in.). The clutch pedal throw (2 - Figure 1) gives the stroke. The lower adjusting screw (2a) limits the pedal throw and thereby the clutch slave cylinder stroke.

Distance B should never be exceeded. The distance is adjusted by removing the fork from the lever and repositioning lever on the cross shaft.

Distance C is set when the clutch is new. During wear of clutch, the distance will decrease. Readjustment should not be needed before it is time to reface the clutch disc.

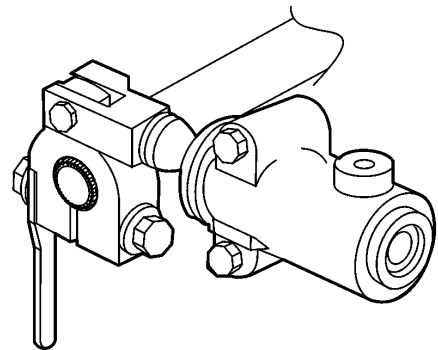
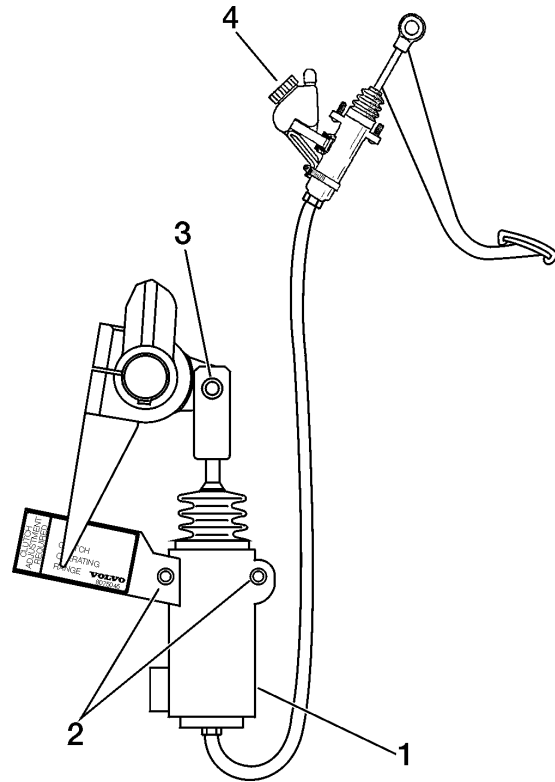


Fig. 2: Volvo Clutch Slave Cylinder

Other Clutch Slave Cylinder

A slave cylinder for any non-Volvo clutch has a wear indicator. When the indicator is out of the operating range, it is time to adjust the clutch.

To get the slave cylinder throw into the accepted range, adjust the clutch plate until the indicator is in the operating range again.



- 1 Slave Cylinder
- 2 Mounting Bolts
- 3 Clutch Clevis Pin
- 4 Clutch Master Cylinder and Reservoir (Found on the Bulkhead in the Engine Coupling)

Power Take-off

Volvo PTO

For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components," publication number 175-001, and appropriate vendor literature.

Volvo PTO's mount directly to the transmission and do not need separate oil fill and check. Make sure there is extra oil filled in the transmission for the PTO volume.



CAUTION

Transmission oil heat exchanger should be installed when using PTO continuously over 15 minutes at a time or with a continuous power output over 55 kW (75 hp). Without heat exchanger, the oil can overheat and transmission damage may follow.

If the application has a remotely installed pump or blower with a driveshaft connection, the driveshaft U-joints need to be greased at every maintenance interval or more often, depending on usage. Use a lithium based grease with EP additives and of the consistency of NLGI No. 2.

Driveshaft

For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components," publication number 175-001, and appropriate vendor literature.

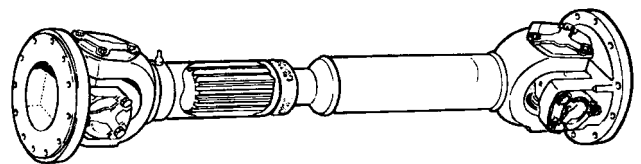
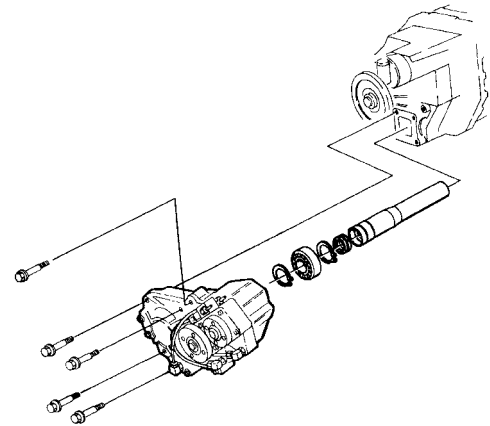
Use a lithium-based grease with EP additives with a consistency of NLGI No. 2. Do not use conventional chassis grease.

The driveshaft U-joints must be lubricated correctly for the bearings to receive grease. The most common case of U-joint failure is incorrect greasing. Always make sure that grease is coming out of all four seals. If one seal fails to purge old grease, move the driveshaft from side to side while applying gun pressure. This allows for greater clearance on the thrust end of the bearing that is not purging. New grease flushes abrasive contaminants from each bearing and assures that the bearing is filled properly.



DANGER

Failure to correctly grease the driveshaft U-joints or slip-joints can lead to component failure which can result in separation of the driveshaft from the vehicle. A separated driveshaft can result in major component damage and loss of vehicle control, and can cause serious personal injury or death.



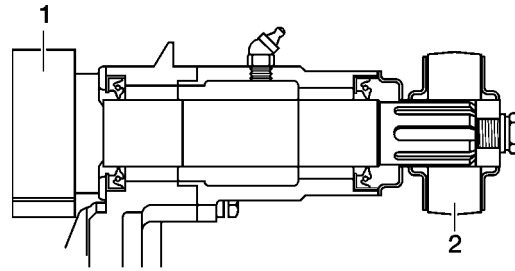
Brakes

Brake Cams and Slack Adjusters

For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components," publication number 175-001, and appropriate vendor literature.

Lubricate the brake cam bushings using a lithium based grease with EP additives and consistency NLGI No. 2. Fill grease until old grease has been pushed out past the seal and new grease can be seen flowing.

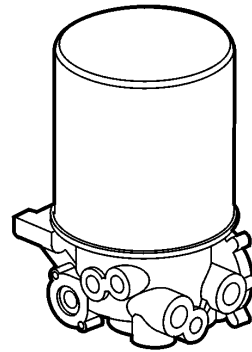
Lubricate the slack adjusters using a lithium based grease with EP additives and consistency NLGI No. 2. Fill grease until old grease has been pushed out past the splined shaft area, and adjustment pawl and new grease can be seen flowing.



Air Dryers

Air dryers have internal maintenance systems that clean out the accumulated moisture frequently and can therefore work with long maintenance intervals. Eventually the drying medium will be filled up and the cartridge will need to be changed. Change cartridge when there is more water being drained in the daily emptying of the tank than usual. This is a progressive development and the time interval will be different from application to application.

Dryer manufacturers recommend changing cartridge every 2 to 3 years but intervals need to be adjusted for application. The only dryer with regular maintenance is the Midland, where the coalescent filter needs changing every year.



Steering System

Steering Linkage and Knuckles

For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components," publication number 175-001, and appropriate vendor literature.

Lubricate the Steering shaft and drag link.

Lubricated the tie rod.

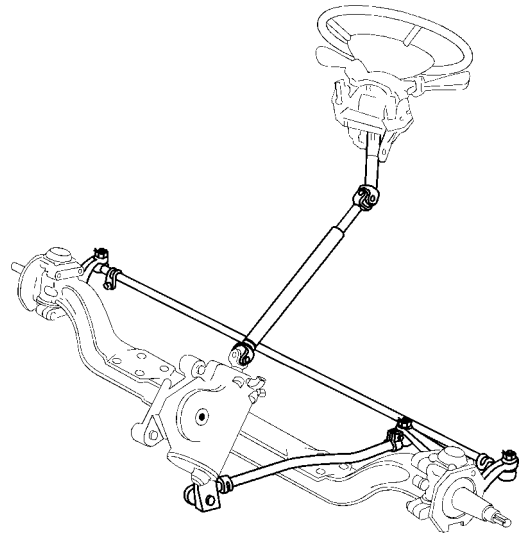
TRW steering gears have a seal at the sector shaft that needs greasing **with a hand gun** every 4 months or more often if the weather and road conditions are severe.

Sheppard steering gears have seals at input shaft and at the sector shaft that need greasing **with a hand gun** every 4 months or more often if the weather and road conditions are severe.

No axial movement is allowable when 100 lbs. of hand pressure is applied. Use hand pressure only, do not check using pliers, wrenches or any other tools.

For all VHD Axle Back models, the steering shaft U-joints should be inspected to see that the plastic caps are intact. Complete this inspection every 4 months. A shorter inspection period interval may be required if the vehicle is operated under severe driving conditions.

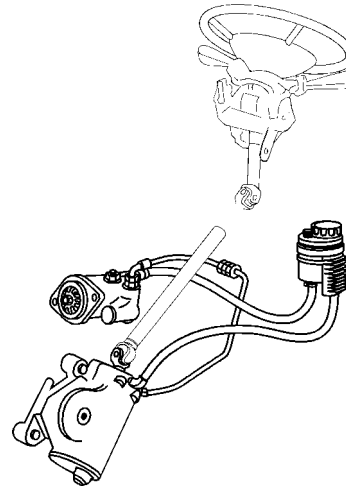
For VHD Axle Forward models, grease the steering shaft U-joints every 4 months. A shorter maintenance interval may be required if the vehicle is operated under severe driving conditions.



Steering System

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A darkened fluid indicates a steering system that is running hotter than normal and the fluid is overheated. Troubleshoot the reason for overheating and change fluid.



Hubs

Oil Lubricated Hubs

For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components," publication number 175-001, and appropriate vendor literature.

Hubs are generally oil lubricated and should be lubricated during maintenance. Change oil when the hub is removed for repairs.



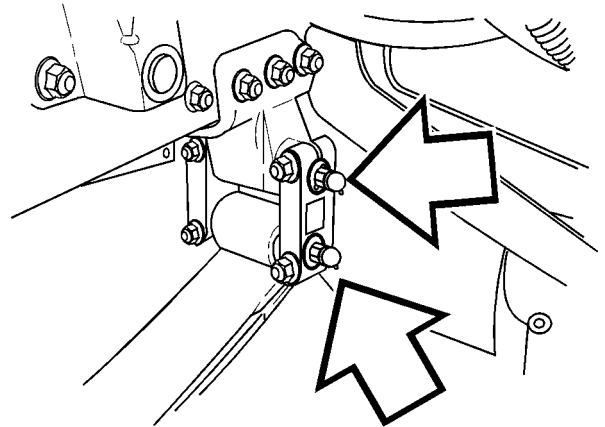
Chassis

Springs and Suspension

Note: For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components, All Models", Publication Number 175-001, and any appropriate vendor literature.

Lubricate spring pins using a lithium based grease with EP additives and the consistency of NLG1 No.2. Fill grease until old grease has been pushed out past the seal on both sides and new grease can be seen flowing. If grease is not flowing through, use a prybar to lever down the spring ends to open up for the grease to flow. To perform this procedure the axle must be free hanging. Refer to the proper Service Publication in Function Group 7.

Wear tolerance for the spring pin and bushing is 5 mm (3/16 in.).



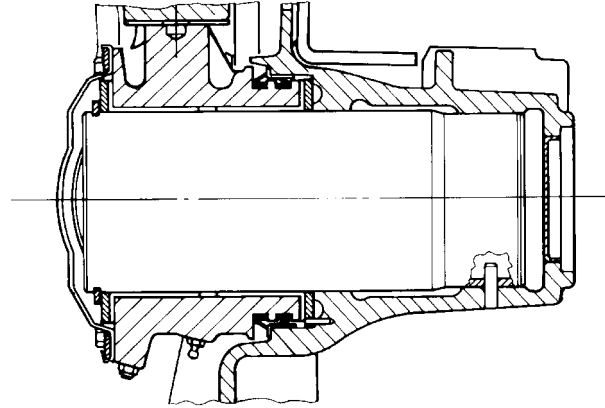
T-Ride Equalizer Beam

Note: For further information concerning component specifications see service information in Group 1, "Oil and Filter Change Intervals for Volvo Components, All Models", Publication Number 175-001, and any appropriate vendor literature.

In regular over-the-road operations the T-Ride should follow the normal lubrication schedule.

It is extremely important that the equalizer beam is well lubricated in any off-road or on/off-road operation. If there are other severe duty constraints, for example driving in water, lubricate more frequently, up to once per day, if necessary.

Note: For further information on the T-Ride Equalizer Beam, Function Group 7, "Rear Suspension, T-Ride".



Cab

Doors

Door lock mechanism should be greased once per year using white lithium grease. Coat the door stop arm with grease every year.



Hood

Hood lock lever is coated with teflon and should not need any conventional lubrication. If the lever is working hard or binds, try cleaning first. Clean and lubricate scissor latch at the base of the cone receptacle with white lithium grease.

